Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential areas of collaboration for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Extreme Environment VCSELs and Drivers for 5G and 6G Systems*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication (√), Advanced Robotics & Manufacturing ( ), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical - Electronics		*Electrical - Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Optical Application Devices	*Laser-related Components and Generation Devices*	*Semiconductor Devices and Systems*	*Other Semiconductor Devices*
Concept of Proposed Technology	High-speed optical links for applications including 5G/6G networks require advances in optical sources and drivers to achieve next-generation performance. Cost-effective and power-efficient systems presently use pulse-amplitude-modulated (PAM-4) vertical-cavity surface-emitting lasers (VCSELs), but the current generation of laser is limited by modulation bandwidth and optical power. The proposed technology focuses on developing high-power, wide bandwidth VCSELs and driver ICs and their application to next-generation short-reach optical interconnects. These VCSELs are engineered to relax fundamental trade-offs among achievable data rate, noise, and signal-to-noise ratio by simultaneously increasing optical output power and extending modulation bandwidth. The higher output power enables the use of digital predistortion to linearize the transmitter and support higher-order modulation formats such as PAM-6 and PAM-8, delivering approximately 50% higher spectral efficiency at a fixed baud rate. In parallel, the enhanced intrinsic bandwidth of the VCSELs			

	supports operation at higher symbol rates, enabling further increases in aggregate throughput. Together, these capabilities provide a scalable path to 200 Gb/s links in the near term, with a clear extension toward 400 Gb/s in future implementations. Compared with silicon-photonics-based approaches, the proposed VCSEL-based solution offers lower cost and reduced system complexity, making it well suited for data center and high-performance computing interconnect applications.
Rationale for Joint R&D Project	The proposed work brings together the Feng, Dallesasse, and Hanumolu Groups. Feng brings experience in extreme-environment, high-speed VCSEL operation. [1-4] Dallesasse brings experience in VCSEL mode control, and in advanced integration. [5-8] Hanumolu brings expertise in high-speed mixed-signal circuit design and is recognized as one of the leading experts in IC design for optical communication systems. [9-14] This joint R&D project is needed to bridge the gap between advanced VCSEL device innovation and deployable high-speed optical interconnect solutions. While the research team will develop the core VCSEL technology, along with the associated circuit and system architectures, close industry collaboration is essential to ensure that these advances translate into manufacturable, commercially viable products. Industrial partners provide critical constraints and feedback on device fabrication capabilities, epitaxial processes, packaging, reliability requirements, and foundry access for the implementation of electronic–photonic integrated circuits (EPIC/EIC). By co-developing the technology within these practical constraints, the project enables early alignment with industry-standard processes and design flows, allowing seamless technology transfer. This collaboration is therefore key to advancing the proposed technology from a mid-level technology readiness level to a high-TRL solution suitable for commercialization, reducing risk and accelerating adoption in real-world data center, 5G/6G and high-performance computing applications.
Keywords	1. Spectrally efficient modulation data form with digital predistortion (200–400 Gb/s links) 2. High-power, wide-bandwidth single mode VCSELs toward 200 Gb/s 3. Extend high-baud-rate short-reach optical interconnects reach to 500 meters. 4. Heterogeneous integration for electronic-photonic integrated circuits.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (4)

R&D Objectives

The UIUC research team has wide-ranging capabilities in the proposed project area, with specific technologies developed within each research group to advance the overall program goals. The technologies from each of the three UIUC research groups involved in the program are described below.

Feng Research Group:

The rapid growth of artificial intelligence, cloud computing and other technologies has led to an unprecedented increase in both data throughput and computational demand. To accommodate the growing demand, data centers are expanding their physical infrastructure and adopting higher-speed interconnects, which extend transmission distances and impose more stringent signal-integrity constraints due to signal attenuation and crosstalk. Ultralow power cryogenic optical data link will be critical solution to achieve lowest (energy/bit), highest bandwidth and capacity and heat reduction in ultrahigh speed data centers. Feng's group has reported the development of 4.5-μm oxide-aperture Cryo-VCSEL with 180 Gb/s PAM-4 data link, showing high-speed operation in extreme environments with multilevel signaling.

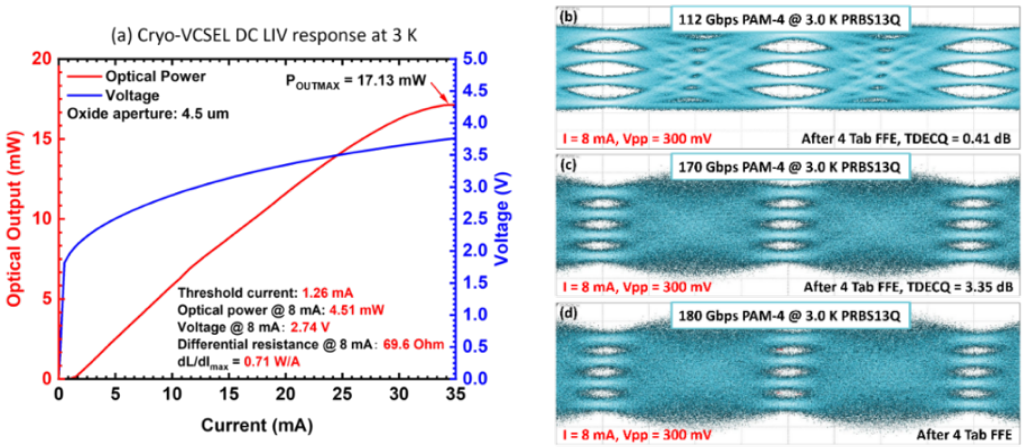

Fig. 1. (a) L-I-V characteristics of a 1/2-λ cavity 4.5 μm aperture Cryo-VCSEL measured at 3.0 K. The device exhibits maximum output power exceeding 17 mW with slope efficiency = 0.71 W/A. (b-d) 3 K Cryo-VCSEL Driven by M8194A: PAM-4 Optical Eye Diagrams at (b) 112 Gbps, (c) 170 Gbps and (d) 180 Gbps. All eye-diagrams are pre-FEC with 4-tap FFE for TDECQ qualification.

Dallesasse Research Group:

The Dallesasse Group has developed methods for mode control for VCSELs that have yielded record single-mode output power and have also been capable of providing single-mode single polarization operation. In addition, heterogeneous integration via epitaxial transfer has been developed by the Dallesasse Group and applied to a variety of photonic devices, including Light-Emitting Transistors (LETs), VCSELs, GaN LEDs, and InAlGaP red lasers.

Work on single-mode lasers began with the application of impurity-induced layer disordering (IILD) to the top distributed Bragg reflector (DBR) to selectively modify top mirror reflectivity. Having a method to pattern the reflectivity across the 2-dimensional plane of the VCSEL aperture allowed suppression of the high-order spatial modes while having a minimal impact on the fundamental mode. In addition, techniques were developed through the introduction of strain in the diffusion mask that allowed control of the shape of the diffusion front under the mask edge. Brought together these allowed the fabrication of VCSELs with record single-mode output power. Such devices allow both preferred coupling characteristics into optical fiber and greater separation in

optical power between energy levels in multi-level signaling applications. Performance of these devices is shown in Fig. 2 below.

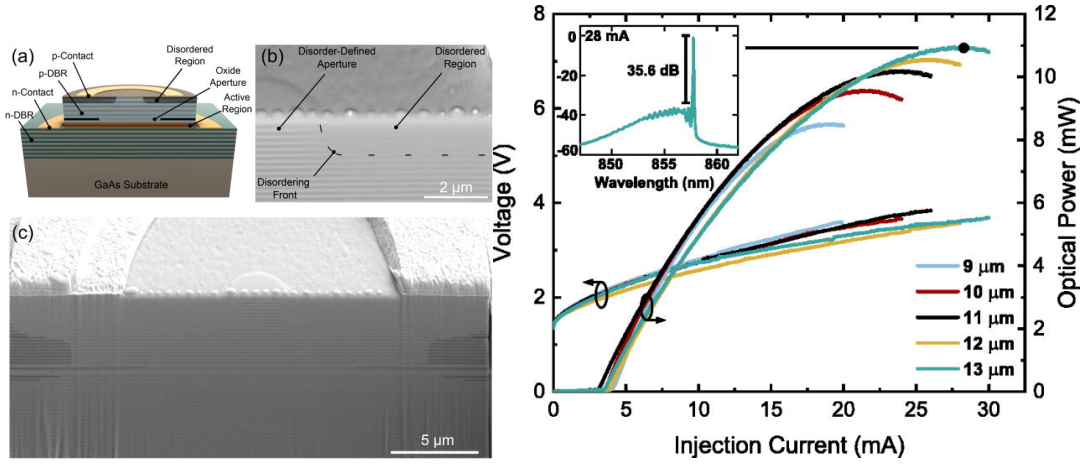

Fig. 2. Schematic diagram of a disordered DBR VCSEL in (a), a scanning electron microscope (SEM) photomicrograph of the diffusion profile in (b), and an SEM photomicrograph of the overall VCSEL cross section in (c). Also shown is the light-current-voltage characteristic illustrating single-mode operation to optical powers greater than 10 mW with a side-mode suppression ratio (SMSR) over 35 dB.

In addition to mode suppression techniques using IILD, the Dallesasse Group has also developed a more manufacturing-friendly method to suppress optical modes using the deposition of high refractive index silicon on the top surface of the VCSEL. This approach allows spatial control of the threshold modal gain by reducing the magnitude of the electric field standing wave on the quantum wells which provide the gain. By creating a silicon annulus, single mode powers near 8 mW have been demonstrated, as shown in Fig. 3 below. Combining IILD and high-index silicon has produced devices with over 13 mW of single-mode power.

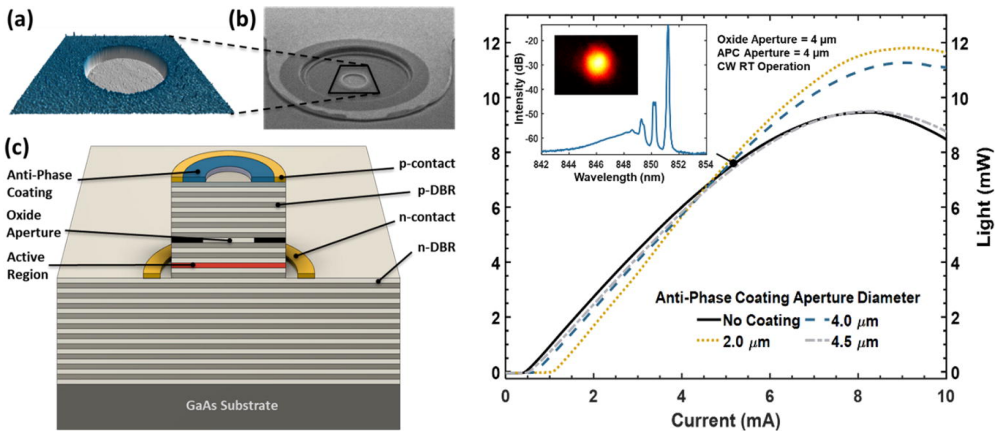

Fig. 3. Atomic force microscope scan in (a), SEM photomicrograph in (b), and diagram in (c) of a single-mode VCSEL using high-index silicon “antiphase” coating. L-I-V and spectra showing single mode performance is also shown.

This technique has been extended through the use of elliptical apertures to break polarization degeneracy and achieve single-mode single-polarization operation. In Fig. 4 below the polarization-resolved L-I-V shows single-polarization operation. In addition to enabling applications in optical systems with polarization-sensitive optical elements such devices are expected to have superior noise performance (RIN) by suppressing polarization switching. Improving noise will further improve the operation of the devices in high-speed communication systems.

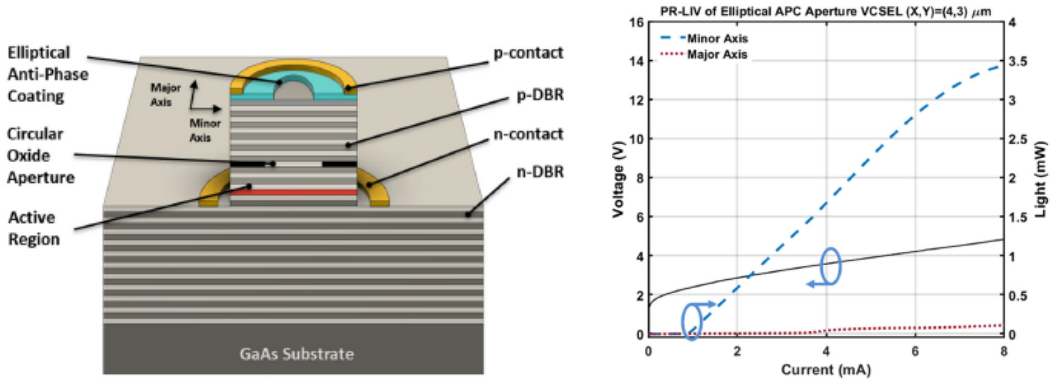

Fig. 4. Schematic and polarization-resolved L-I-V characteristic of VCSEL incorporating an elliptical-aperture mode-suppression filter.

Finally, the Dallesasse Group has developed epitaxial transfer methods to allow the heterogenous integration of VCSELs on silicon. In this method VCSELs are fabricated after the transfer process allowing precise registration of the VCSEL aperture relative to features on the silicon host wafer. An additional advantage is superior operation over temperature relative to devices fabricated on native substrates. This is provided by the higher thermal conductivity of silicon relative to GaAs. These devices are shown in Fig. 5 below.

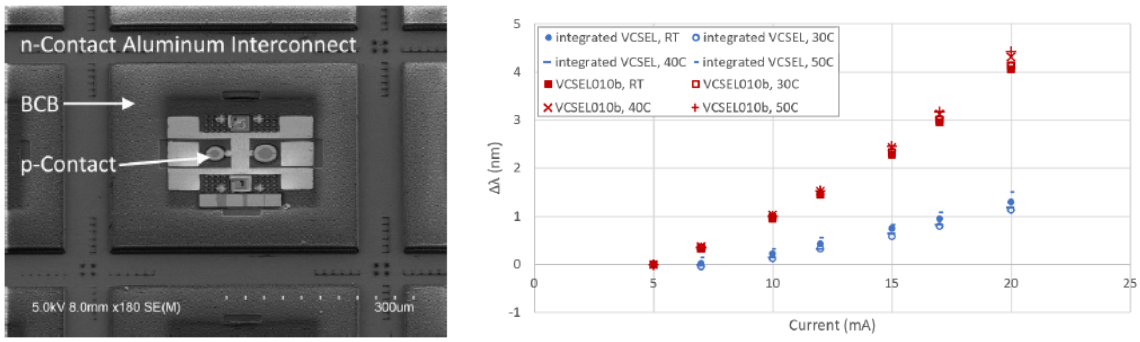

Fig. 5. SEM photomicrograph of VCSEL integrated on silicon, and data showing the wavelength change with increasing drive current for VCSELs integrated on silicon versus VCSELs on their native GaAs substrate.

Hanumolu Research Group

The Hanumolu Group has developed a strong foundation of circuit- and system-level technologies that directly support the objectives of this R&D project. These capabilities include high-swing, wide-bandwidth driver architectures, 100 Gb/s serializer designs, DAC-based digital predistortion techniques for transmitter linearization, and ultra-low-jitter clocking solutions for high-speed optical links. The project will leverage these existing technologies and the group's established expertise to accelerate development of the proposed VCSEL-based transmitters, while further advancing their performance and scalability toward 200–400 Gb/s operation.

R&D Target Specifications

Target specifications are defined for VCSEL and link performance. VCSEL performance is measured at the device level using high-speed test equipment, direct device probing, and high-speed receivers. Link tests are performed with VCSEL being electrically driven by the driver IC. Industry-standard methods are used for measurement of device performance parameters. Where system performance parameters have test methods defined by bodies such as the IEEE 802.3 working group those are used for verifying compliance. Anticipated VCSEL specifications are presented in Table 1 below and corresponding system performance in Table 2.

Table 1: VCSEL Performance

Parameter	Target Specification	Notes
Emission Wavelength	830 to 860 nm	
SMSR	-30 dBm	
Optical Bandwidth	28 GHz (RT) 112 Gb/s PAM4 56 GHz(77K) 224 Gb/s PAM4 84 GHz(77K) 336 Gb/s PAM4	149 Gb/s PAM6 298 Gb/s PAM6 448 Gb/s PAM6
RIN	< -120 dB/Hz	
Threshold Current	1 to 2 mA	
Slope Efficiency	>0.5 W/A	
Peak Output Power	>4 mW (RT), >10 mW (77K)	

Table 2: Link Performance

Parameter	Target Specification	Notes
Link Data Rate	200 Gb/s (near term), scalable to 400 Gb/s	Single-wavelength IM-DD link
Modulation Format	PAM-4 /PAM-6/PAM-8	PAM-8 enabled via DPD and high optical power
Baud Rate	50–150 GBd	Enabled by wide VCSEL and driver bandwidth
Modulation Bandwidth	≥ 28–85 GHz	Small-signal 3 dB bandwidth
Optical Output Power	≥ 10 dBm	Supports higher-order modulation and SNR margin
TX Driver Output Swing	≥ 2 Vpp differential	High-swing, wideband driver architecture
TX Predistortion	DAC-based digital predistortion	Compensates VCSEL nonlinearity and bandwidth limits
TX Clock Jitter	< 50 fs rms (integrated)	Critical for PAM-8 eye integrity
Receiver Type	Linear TIA + broadband photodiode	Optimized for high-speed IM-DD
RX Bandwidth	≥ 40–80 GHz	Matches highest baud-rate operation
Link Reach	500 m (short reach)	Data center / HPC interconnects
Implementation	VCSEL + CMOS EIC	Compatible with volume manufacturing

R&D Plan

The tasks and milestones below will be achieved using design, fabrication, and test infrastructure available at UIUC. VCSEL epitaxial material will be sourced through commercial epitaxial wafer suppliers.

Tasks and Milestones

Task 1.0: Design VCSEL epitaxial layer structure optimized for speed and power
Task 2.0: Design VCSEL device geometry, mask levels, and process flow
Task 3.0: Create preliminary VCSEL equivalent circuit models for IC design
Task 4.0: Preliminary design of driver IC
Task 5.0: Fabricate VCSELs incorporating mode control structures on native (GaAs) substrates
Task 6.0: Refine VCSEL electrical model based on measured results
Task 7.0: Final design of driver IC
Task 8.0: Demonstrate epitaxial transfer to create VCSELs on silicon
Task 9.0: Test Driver ICs and VCSELs
Task 10.0: Heterogenous integration of VCSELs with driver ICs
Task 11.0: Test link performance

Milestone 1.0: VCSEL epitaxial wafers procured
Milestone 2.0: VCSELs fabricated on native substrates
Milestone 3.0: VCSELs fabricated on silicon
Milestone 4.0: Driver ICs out of fab
Milestone 5.0: VCSELs fabricated on driver ICs
Milestone 6.0: Optical link demonstrated
Anticipated task and milestone timing is presented in the Gantt chart below.

Gantt Chart

	2027				2028				2029	
Task/Milestone	**Q1**	**Q2**	**Q3**	**Q4**	**Q1**	**Q2**	**Q3**	**Q4**	**Q1**	**Q2**
Task 1.0	X									
Task 2.0	X	X								
Task 3.0		X								
Task 4.0	X	X	X							
Task 5.0			X	X						
Task 6.0				X						
Task 7.0				X	X					
Task 8.0			X	X	X	X				
Task 9.0							X	X		
Task 10.0							X	X	X	
Task 11.0									X	X
Milestone 1.0			X							
Milestone 2.0				X						
Milestone 3.0						X				
Milestone 4.0							X			
Milestone 5.0									X	
Milestone 6.0										X

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Provide design expertise in advanced VCSEL structures, including single mode, single polarization, and extreme environments. • Fabricate VCSELs using III-V device fabrication infrastructure available in UIUC's Holonyak Micro and Nanotechnology Laboratory. • Provide design expertise in high-speed laser driver design. • Develop methods for heterogeneous integration of VCSELs on silicon ICs for speed and efficiency improvements. • Provide test expertise and infrastructure for high-speed characterization of devices. • Provide test expertise and infrastructure for characterization of Tx system, including laser driver. • Perform DC, RF, PAM-4, and PAM-6 datalink tests. • Show scalability path for multi wavelength arrays capable of 4X400 Gbps (1.6 Tbps).
Expected R&R of Korean SME Partner	• Participate in regular technical reviews, joint design iterations, and knowledge exchange to ensure close alignment between academic research objectives and industry needs. • Provide domain expertise in VCSEL fabrication, packaging, reliability, and high-volume manufacturing constraints to guide device, circuit, and system co-design decisions. • Advise on productization requirements, cost targets, and supply-chain considerations to ensure alignment with commercial deployment. • If capable, provide MOCVD growth for VCSEL structures designed at UIUC. • If capable, provide IC shuttle run access for laser driver designs created at UIUC. • Provide system integration expertise for optical transceiver applications. • Provide commercial insight on emerging applications and resulting performance needs.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Reduction power integrity for CPO high speed interconnect. • Cryo-VCSEL Bandwidth > 50 GHz (77K) and 140 Gb/s PAM4 and 180 Gbps PAM6 @ low bias I < 4 mA • Co-designed VCSEL devices, high-swing driver circuits, and DAC-based digital predistortion achieving spectrally efficient PAM4/PAM-6 transmission at high baud rates. • Integrated transmitter and receiver architectures validated under realistic industry constraints, including packaging, and power efficiency. • Advancement of VCSEL-based optical links from mid-TRL laboratory prototypes to high-TRL, system-level demonstrators. • Reduced technology transfer risk through early industry engagement, enabling smoother transition to volume production. • Creation of a low-cost alternative to silicon-photonics-based short-reach transceivers for data center and HPC markets.

6. References

1. Haonan Wu, Wenning Fu, Yulin He, Zetai Liu, and Milton Feng “Laser Cavity Temperature and 3D Nonlinear Thermal Model of VCSEL from 2.6 to 130 K," IEEE Journal of Quantum Electronics, VOL. 61, No.6, 2400512 page 1-12 (Dec. 2025). Print ISSN: 0018-9197, Online ISSN: 1558-1713, Digital Object Identifier (DOI): 10.1109/JQE.2025.3612272
2. Derek Chaw, Haonan Wu, Zetai Liu, and Milton Feng, “Oxide Aperture Scaling of Cryogenic Microcavity Surface Emitting Laser for Ultrafast Optical Link,” Journal of Select Topic in Quantum Electronics (JSTQE), vol.31, No.2, Paper 1701009, March /April 2025
3. Haonan Wu, Derek Chaw, Zetai Liu, Yulin He, and Milton Feng “Sub-micron aperture VCSEL demonstrates ultralow I_{TH} =0.05 mA and energy/bit = 45.5 fJ/bit at 3K,” Appl. Phys. Lett. (126), 181104 (2025) – (Feng group)
4. Zetai Liu, Haonan Wu, Yulin He, Derek Chaw, and Milton Feng, “180 Gb/s PAM-4 Optical Link by Cryogenic VCSEL,” to be presented in 2026 Optical Fiber Comm. (OFC26)
5. P. Su, K.P. Pikul, M.D. Kraman, and J.M. Dallesasse, “High-power single-mode vertical-cavity surface-emitting lasers using strain-controlled disorder-defined apertures,” Appl. Phys. Lett. 119, 241101 (2021), https://doi.org/10.1063/5.0068713. (Editor’s Pick)
6. K.P. Pikul, P. Su, M.D. Kraman, B.A. Kesler, and J.M. Dallesasse, “Utilization of silicon optical coatings for transverse-mode suppression in high-power oxide-confined vertical-cavity surface-emitting lasers,” Appl. Phys. Lett. 126, 201107 (2025), doi: 10.1063/5.0260091.
7. K. Pikul, L. Espenhahn, P. Su, M. Kraman, J.M. Dallesasse, “Polarization Control in Vertical-Cavity Surface-Emitting Lasers via Elliptical Aperture Definition in Optical Coatings,” CS MANTECH 2024, May 20-23, Tucson, AZ (2024).
8. Leah Espenhahn, John Carlson, Patrick Su, John Dallesasse, "Heterogeneously integrated VCSELs on silicon," Proc. SPIE 12020, Vertical-Cavity Surface-Emitting Lasers XXVI, 120200J (2 March 2022); https://doi.org/10.1117/12.2608839.
9. E. Abdelrahman, M. Badr Younis, M. Osama Selim, M. Saad Aly, M. A. Khalil and P. Hanumolu, "A 1.54-pJ/b 64-Gb/s 16-QAM Intradyne Coherent Optical Receiver in 28-nm CMOS," in *IEEE Journal of Solid-State Circuits*, vol. 61, no. 1, pp. 20-32, Jan. 2026.
10. A. E. Abdelrahman, M. G. Ahmed, M. A. Khalil, M. Badr Younis, K. -S. Park and P. Hanumolu, "A Carrier Phase Recovery Loop for a 3.2-pJ/b, 24-Gb/s QPSK Coherent Optical Receiver," in *IEEE Journal of Solid-State Circuits (to appear).*
11. Y. Li *et al.*, "A −11.6-dBm OMA Sensitivity 0.55-pJ/bit 40-Gb/s Optical Receiver Designed Using a 2-Port-Parameter-Based Design Methodology," in *IEEE Open Journal of the Solid-State Circuits Society*, vol. 4, pp. 328-339, 2024.

12. M. G. Ahmed, D. Kim, R. K. Nandwana, A. Elkholy, K. R. Lakshmikumar and P. Hanumolu, "A 16-Gb/s -11.6-dBm OMA Sensitivity 0.7-pJ/bit Optical Receiver in 65-nm CMOS Enabled by Duobinary Sampling," in *IEEE Journal of Solid-State Circuits*, vol. 56, no. 9, pp. 2795-2803, Sept. 2021.
13. M. A. Khalil *et al*., "A 69.3fs Ring-Based Sampling-PLL Achieving 6.8GHz-14GHz and −54.4dBc Spurs Under 50mV Supply Noise," *2024 IEEE International Solid-State Circuits Conference (ISSCC)*, San Francisco, CA, USA, 2024.
14. M. G. Ahmed *et al*., "A 12-Gb/s -16.8-dBm OMA Sensitivity 23-mW Optical Receiver in 65-nm CMOS," in *IEEE Journal of Solid-State Circuits*, vol. 53, no. 2, pp. 445-457, Feb. 2018.